

李金城

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教育背景

香港中文大学	09/2024-10/2025	
理学硕士	专业: 机械与自动化工程	GPA: 3.54/4.0
底特律大学	08/2023-06/2024	
工学学士	专业: 机器人与机电系统工程	GPA: 3.91/4.0
北京化工大学	09/2020-06/2024	
工学学士	专业: 机械设计制造及其自动化(机器人方向)	GPA: 3.33/4.33

论文发表

李金城, 曹乐涵, 陈剑锋 (共同第一作者), *Working principle and application analysis of UART*, 被 2022 年第四届国际电子、机电与自动化会议 (ICEMA 2022) 录用, 收录至 IEEE Xplore, DOI: 10.1109/EEBDA56825.2023.10090571

李金城, *Working Principle and Application Analysis of Phase-locked-Loop*, 被国际机电与智能系统会议 (CONF-MSS 2023) 录用, DOI: 10.54254/2755-2721/11/20230228

校园经历

香港中文大学毕业设计
四电机线驱动 2D 机器人 08/01/2025-30/4/2025

- 从零搭建了一个 2D 四电机线驱动机器人平台, 实现了末端执行器在二维平面内的精确移动
- 使用树莓派 4B 作为主控, 利用加装的 Waveshare 2-CH CAN FD HAT 拓展板将树莓派的 SPI 接口转化为 CAN 总线接口, 与四块 VESC FSESCv6.7 驱动板相连, 利用 CAN 总线实现驱动板与树莓派的通信, 进而控制四路电机。再配合根据末端执行器位置逆解出的关节轨迹, 实现了对末端执行器位置和轨迹的精确控制
- 在此毕业设计中使用 Python 编写了树莓派与 VESC 驱动板配合的函数库, 包括角度控制、电流控制和转速控制等基础函数, 以及逆解末端执行器位姿, 关节轨迹规划等辅助代码

香港中文大学
高级机器人学, Bruce 双足机器人平衡控制与动作设计 05/01/2025-11/5/2025

- 实现了 Bruce 机器人的双足稳定站立, 以及行走、蹲起、跳跃等基础动作
- 使用 QP 算法确保机器人在执行不同动作时姿态的动态平衡, 以及在质心转移过程中的姿态平衡
- 编写了能够针对不同任务规划质心的轨迹, 并下发到底层实现机器人的各种动作中层代码
- 编写了面向用户端的顶层代码, 能够获取用户输入并下发到中层代码实现用户对机器人的操控以及多任务切换

底特律大学毕业设计
自主式物品取送机器人系统 06/01/2024-22/4/2024

- 与两位同学共同搭建了一个融合计算机视觉, 机械臂抓取和多目标点导航的室内辅助机器人系统, 可以通过语音交互获得用户的需求, 实时识别并抓取相应的物品并送至用户手中
- 机器人使用 Pioneer 3DX 作为平台, 通过 Move_Base 功能包进行导航, 微软 Azure 引擎实现语音交互功能, Kinect2 深度相机和 YOLOv8 实现物品识别, Trossen RX200 机械臂与 Moveit 实现机械臂的训练与抓取
- 我负责多目标点导航部分的实现, 包括 Move_Base 中全局规划器和局部规划器的选择和调参; Gmapping 建图以及代价地图参数调整; 使用 A*算法编写多目标点导航节点以及与机械臂和视觉模块通信节点的编写

底特律大学
机器人学, Trossen RX200 机械臂控制与仿真 02/02/2024-16/4/2024

- 仿照 Trossen RX200 机械臂, 使用 Solidworks 重新建模出了 RX200 的模型, 并将其导入 MATLAB 中使用 AppDesigner 工具包进行后续的功能设计和控制, 实现了机械臂在 Matlab 中的仿真以及利用 ROS 实时控制
- 使用几何法/DH 参数法构建出了机器人各部分/各关节的正向运动矩阵
- 通过这段经历, 掌握了机器人运动学正逆解的相关知识, 以及如何使用 MATLAB 编写 ROS 节点

个人优势

- 本科期间为 3+1 中美联合培养, 有一年的美国交换经历, 英语口语流利
- 熟练使用 PID, LQR 等经典控制模型, 有丰富的 Sim2Real 经验
- 有丰富的 Python, C++ 开发经验, 以及德州仪器, STM 公司单片机开发经验
- 熟练掌握 Linux 系统下 ROS 相关的基础开发

Jincheng Li

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Education

The Chinese University of Hong Kong

New Territories, Hong Kong, CN

MSc in Mechanical Automation Engineering, GPA: 3.54/4.0

Sep 2024 - Oct 2025

University of Detroit Mercy

Detroit, MI, US

B.E. in Mechatronics, Robotics, and Automation Engineering, GPA: 3.91/4.00

Aug 2023 – Jun 2024

Beijing University of Chemical Technology

Chaoyang, Beijing, CN

B.E. in Mechanical Design, Manufacturing and Automation, GPA: 3.33/4.0

Sep 2020 – Jun 2024

Publication

Li Jincheng, Cao Lehan, Chen Jianfeng (co-first authors): Working Principle and Application Analysis of UART, accepted by the 4th International Conference on Electronics, Electromechanics, and Automation (ICEMA 2022), indexed in IEEE Xplore, DOI: 10.1109/EEBDA56825.2023.10090571.

Li Jincheng, Working Principle and Application Analysis of Phase-locked-Loop, accepted by the International Conference on Mechatronics and Smart Systems (CONF-MSS 2023), DOI: 10.54254/2755-2721/11/20230228.

Project

The Chinese University of Hong Kong

Four-Motor Cable-Driven 2D Robot (Graduation Project)

Jan 2025 – Apr 2025

- Built a 2D four-motor cable-driven robot platform from scratch, achieving precise movement of the end-effector within a 2D plane.
- Used a Raspberry Pi 4B as the main controller, equipped with a Waveshare 2-CH CAN FD HAT board to convert the Raspberry Pi's SPI interface to a CAN bus interface for communication with four VESC FSESCv6.7 motor driver boards. Leveraged the CAN bus to enable communication between the motor drivers and the Raspberry Pi, achieving precise control of the end-effector position and trajectory through inverse kinematics.
- Developed a Python library for Raspberry Pi to interact with VESC drivers, including basic functions such as angle control, current control, and speed control, as well as auxiliary code for inverse kinematics and trajectory planning.

The Chinese University of Hong Kong

Jan 2025 – May 2025

Advanced Robotics: Bruce Biped Robot Balance Control and Motion Design

- Achieved stable biped standing for the Bruce robot, as well as basic motions such as walking, squatting, and jumping.
- Ensured dynamic balance during motion execution using QP algorithms and maintained posture stability during center-of-mass shifts.
- Designed trajectory planning for the robot's center of mass to execute various actions, and implemented mid-level motion control code.
- Developed user-facing top-level code to allow user input for controlling the robot and switching between multiple tasks.

University of Detroit Mercy

Autonomous Object Delivery Robot System (Graduation Project)

Jan 2024 – Apr 2024

- Collaborated with two teammates to develop an indoor assistive robot system combining computer vision, robotic arm manipulation, and multi-goal point navigation. The system could understand user requirements via voice interaction, recognize and grab specific objects in real time, and deliver them to the user.
- Utilized Pioneer 3DX as the base platform, Move_Base for navigation, Microsoft Azure for voice interaction, Kinect2 with YOLOv8 for object recognition, and Trossen RX200 robotic arm with MoveIt for robotic arm training and manipulation.
- Took charge of implementing multi-goal navigation, including selecting and tuning global and local planners in Move_Base, Gmapping for mapping, and cost map parameter adjustments. Developed navigation nodes using the A* algorithm for multi-goal navigation and created communication nodes for integration with the arm and vision modules.

University of Detroit Mercy

Robotics: Trossen RX200 Robotic Arm Control and Simulation

Feb 2024 – Apr 2024

- Recreated the Trossen RX200 robotic arm model using SolidWorks and imported it into MATLAB for subsequent functionality design and control using the AppDesigner toolkit. Achieved robotic arm simulation in MATLAB and real-time control via ROS.
- Constructed forward kinematics matrices for the robot components and joints using geometric methods/DH parameters.
- Gained expertise in robotic kinematics (forward and inverse solutions) and developed ROS nodes using MATLAB.

Personal Strengths

- Participated in a 3+1 Sino-American joint program during undergraduate studies, with one year of exchange experience in the USA. Fluent in spoken English.
- Proficient in classical control models such as PID and LQR, with extensive Sim2Real experience.
- Experienced in Python and C++ development, as well as microcontroller development with Texas Instruments and STM.
- Skilled in ROS development within Linux systems.